



HealthConnect Model Handoff - Full Exchange with ML Engineering

Week 8 - Data Science Track

AnalystLab Africa - Experience Lab Internship Programme

This document is the complete, chronological record of the model handoff exchange with ML Engineering kept as one file so the full cycle (request → review → fix → follow-up) is traceable in a single place, rather than split across separate messages.

1 - Initial Request (Data Science → ML Engineering)

I have a finalized no-show prediction model from Week 7 (Random Forest, ROC-AUC ≈ 0.67 - 0.68 on unseen patients) with a documented spec (input features, output format, recommended threshold 0.30 - 0.35 , and a known limitation: an unresolved gender fairness gap).

Could you let me know: (1) what input/output format your pipeline expects, and (2) whether there are any technical constraints on my side (e.g. serialization format, feature types) that I should account for before we finalize the handoff? Happy to share my full model spec doc. Thanks!

2 - ML Engineering's Response

ML Engineering *Week 8*

Input format:

gender	(str)	-- "Female" / "Male"
age	(int)	
age_group	(str)	-- e.g. "35-44"
appointment_type	(str)	-- e.g. "Follow-up", "Specialist Consultation"
appointment_day	(str)	-- e.g. "Tuesday"
appointment_time	(str)	-- e.g. "Afternoon"
booking_lead_days	(int)	
previous_appointments	(int)	
previous_no_shows	(int)	
reminder_sent	(str)	-- "Yes" / "No"
reminder_channel	(str)	-- "SMS" / "WhatsApp" / "Email" / "None_Sent"
distance_to_clinic_km	(float)	

Output format:

```
{
  "appointment_id": "HC-00412",
  "no_show_probability": 0.5628,
  "risk_tier": "MODERATE_RISK",
  "recommended_action": "TRIGGER_SMS_REMINDER_REQUESTING_CONFIRMATION"
```

```
}
```

Technical constraints:

- Serialization must be pickle-compatible (scikit-learn .pkl)
- waiting_time_minutes must NOT be included (data leakage)
- Categoricals must come in as strings, not integers

3 - Follow-Up (Data Science → ML Engineering)

Thanks again for catching that, I've now fixed it and re-verified on the real dataset, wanted to close the loop quickly.

I retrained the model without waiting_time_minutes and re-ran the full evaluation. Good news: performance is virtually unchanged (ROC-AUC 0.682, same as before to 3 decimal places), which confirms that feature wasn't actually doing much for the model so removing it cost nothing. The recommended threshold (0.30–0.35) and the known limitations (the lead-time blind spot and the gender fairness gap) are all unaffected.

One thing DID change, and I want to flag it rather than bury it: when I redid the baseline comparison with the leakage feature properly removed from both models, the improvement over the Week 5 baseline is no longer statistically significant at 95% (bootstrap CI now includes zero, versus excluding it before). The point estimate still favours the current model, but I'm revising my claim from statistically real improvement to mall, favourable, but not conclusive didn't want to quietly keep the more favourable framing now that it's not accurate.

Still working on packaging the preprocessing + model into a single pickle-compatible pipeline that accepts raw string categoricals directly, per your spec that part is still pending, separate from the leakage fix. Will follow up once that's ready. Thanks again for the thorough review it made the final model more honest, not just cleaner.

4 - Status Summary

Resolved: data leakage feature removed and re-verified; recommended threshold and risk-tier mapping unchanged.

Revised (reported, not hidden): baseline-improvement significance claim, now more conservative.

Pending: packaging preprocessing and model into a single pickle-compatible pipeline accepting raw string categoricals. Full detail in the companion document HealthConnect_Model_Specification.docx.